SHS Lab Test — 2026-06-29

Clark South Ballast Tank, WHOI

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Test date: 2026-06-29. The data record also includes the 2026-06-30 plastic-wick follow-up and post-test data through 2026-07-01 16:18 UTC.

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1 Setup

The test repeated the 2026-05-28 dunk-cycle workflow with two configurations. On 2026-06-29 the SHS was run in the cotton wick setup; the fan was raised to maximum after the first OUT event. The 2026-06-30 follow-up changed to a plastic friction-fit wick at 12:55 UTC, with the fan turned on at the first IN.

Instrument	ID	Rate
RBRduo3 T.T (dry/wet bulb)	SN 233860	3 s
Vaisala WXT520 (met)	ASWXT203	1 min
Tank CT reference	—	1 Hz nominal

Event Log

Cycle	IN	OUT	Duration
J29-IN-1	06-29 15:28	15:33	5 min
J29-IN-2	06-29 16:03	16:10	7 min
J29-IN-3	06-29 16:37	16:40	3 min
J29-IN-4	06-29 17:18	17:23	5 min
J29-IN-5	06-29 17:58	18:04	6 min
J29-IN-6	06-29 18:40	18:44	4 min
J29-IN-7	06-29 19:15	19:20	5 min
J29-IN-8	06-29 19:56	19:59	3 min
J30-IN-1	06-30 13:08	13:14	6 min
J30-IN-2	06-30 13:46	13:50	4 min
J30-IN-3	06-30 14:23	14:29	6 min
J30-IN-4	06-30 14:58	15:30	32 min
J30-IN-5	06-30 16:02	16:08	6 min

Ambient Conditions

Phase	T (°C)	RH (%)	p (mbar)	wind (m/s)
2026-06-29 cotton wick	21.33	72.5	1016.82	0.94
2026-06-30 plastic wick	21.21	72.9	1018.09	0.76

2 Source Notes and Photos

Testing Notes

The testing notes are reflected in the setup description and event log above. They record RBR/WXT startup at 14:50 UTC, fan level 2 at 15:23 UTC, tank CT logging at 15:27 UTC, the first SHS dunk at 15:28–15:33 UTC, and the fan increase to maximum after the first OUT event with both probes facing the fan. For the follow-up, the notes record the 12:55 UTC change to the plastic friction-fit wick and the five subsequent dunk cycles.

3 Humidity: WXT520 vs. Psychrometric

Phase	N	WXT RH	psych RH	ΔRH	Δe
2026-06-29 cotton wick	263	72.47	74.23	+1.75	+0.52
2026-06-30 plastic wick	273	72.90	93.57	+20.67	+5.34

The cotton-wick day tracks the WXT520 closely during active in-air windows. The plastic-wick day shows a much larger psychrometric humidity offset, so that configuration should be treated as unresolved until the wick geometry and wetting state are checked. The full post-test record remains useful for seeing wick drying and long-term drift, but the table above uses only active in-air samples from the note-defined test windows.

4 Temperature and Tank Reference

Cycle	Phase	T_{tank}	$T_d - T_{\text{tank}}$	$T_w - T_{\text{tank}}$	N
J29-IN-1	29 cotton wick	20.810	+0.006	+0.001	101
J29-IN-2	29 cotton wick	20.805	+0.003	-0.014	141
J29-IN-3	29 cotton wick	20.799	+0.043	-0.273	61
J29-IN-4	29 cotton wick	20.794	+0.077	-0.375	101
J29-IN-5	29 cotton wick	20.788	+0.003	-0.018	121
J29-IN-7	29 cotton wick	20.783	+0.113	-0.437	101
J30-IN-1	30 plastic wick	20.517	+0.041	+0.040	121
J30-IN-2	30 plastic wick	20.517	+0.018	+0.006	81
J30-IN-3	30 plastic wick	20.513	+0.053	-0.024	121
J30-IN-4	30 plastic wick	20.511	+0.009	-0.017	641
J30-IN-5	30 plastic wick	20.511	+0.115	+0.105	121

5 Response Times

Response-time estimates are computed separately for each OUT event and variable. For each segment, time is re-zeroed at OUT ($t = 0$) and data are fit with

$$y(t) = A + B\, e^{-t/\tau}$$

over the interval from OUT to min(OUT + 25 min, next IN). The reported τ values are medians across events in each phase. If the optimizer reaches the upper bound ($\tau = 7200$ s), the result



Figure 1: Setup photographs from `data/20260629Lab`. Top: June 29 cotton-wick layout with the WXT520, SHS wet/dry-bulb mast, fan, and ballast tank. Bottom: June 30 plastic-wick follow-up, including the friction-fit wick close-up and wider fan-facing instrument layout.

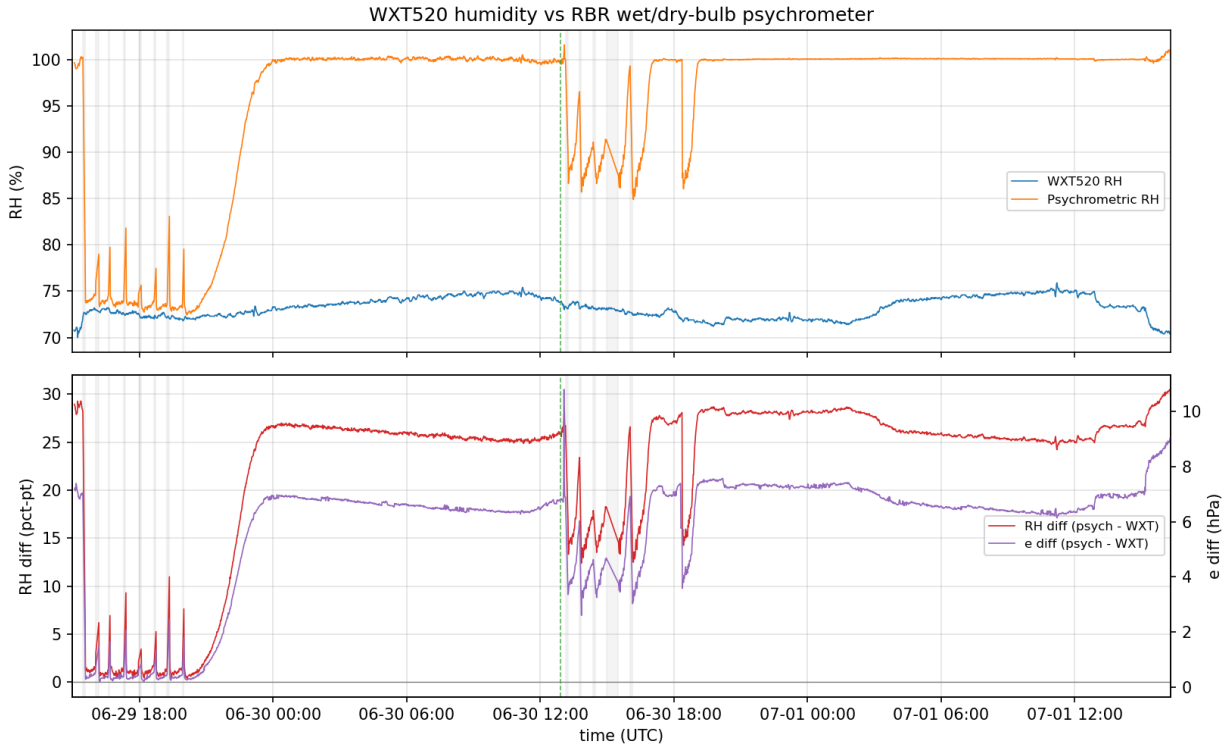


Figure 2: Top: WXT520 RH and RBR wet/dry-bulb psychrometric RH. Bottom: psychrometric minus WXT RH difference and vapor-pressure difference. Grey bands mark in-water periods; the green dashed line marks the June 30 plastic-wick change.

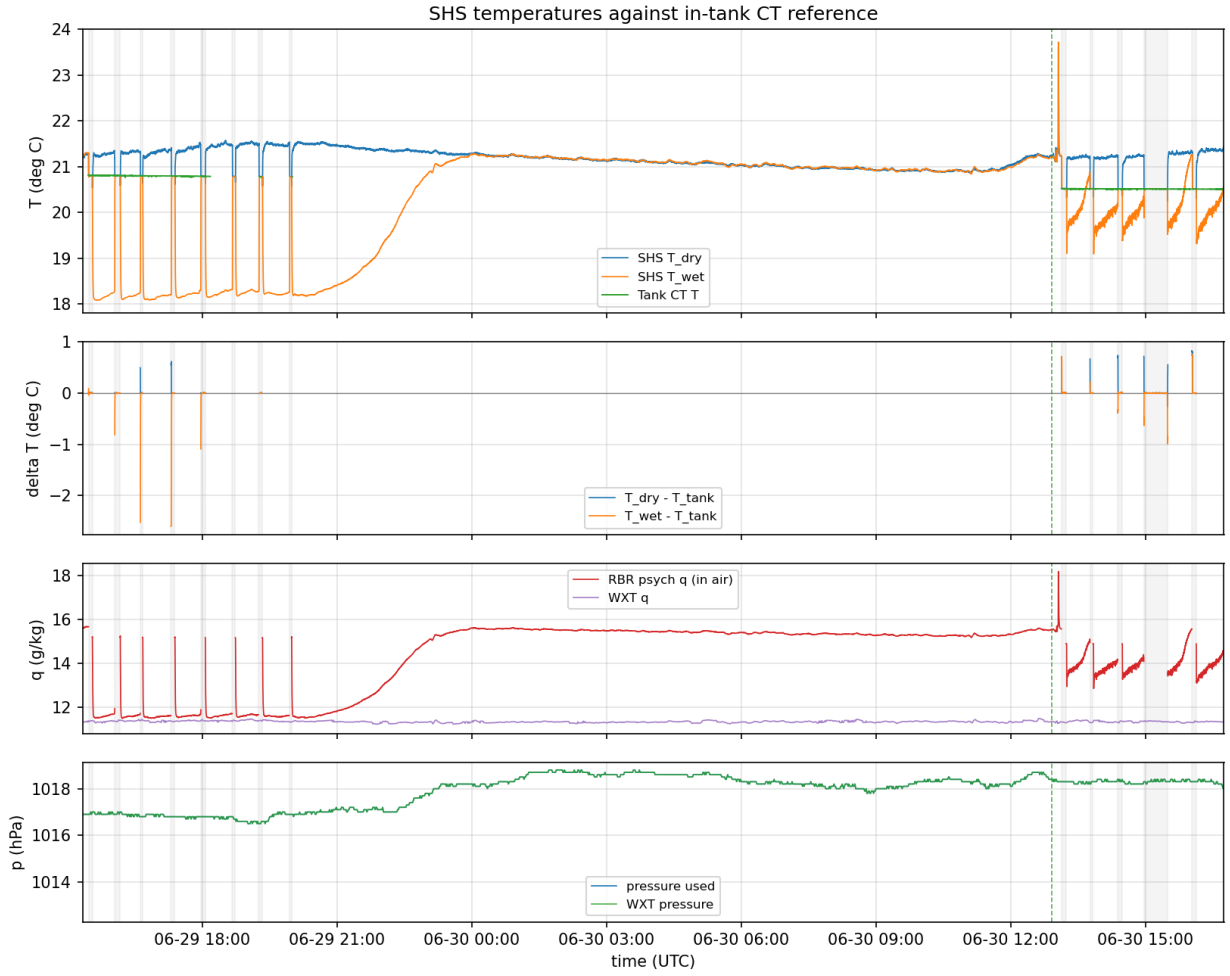


Figure 3: From top to bottom: SHS dry- and wet-bulb temperatures with the in-tank CT reference; SHS minus tank-temperature residuals during periods with CT coverage; WXT and in-air RBR psychrometric specific humidity; and air pressure used in the psychrometric calculation with WXT pressure. Grey bands mark in-water periods; the green dashed line marks the June 30 plastic-wick change.

is marked as capped (non-exponential or unresolved) rather than interpreted as a physical time constant.

The settling metrics in the second table are computed from the wet-bulb curve: the wet-bulb series is smoothed (7-point centered rolling mean), the cooling-rate peak is the minimum of dT_w/dt, and t_{knee} is the first time after that peak when dT_w/dt rises above 10% of the peak-cooling rate. The 63%-response time is $t_{63} = \min\{t : T_w(t) \le T_{w,0} - 0.63\,(T_{w,0} - T_{w,\text{plat}})\}$, where $T_{w,\text{plat}}$ is the plateau estimate used for each interval.

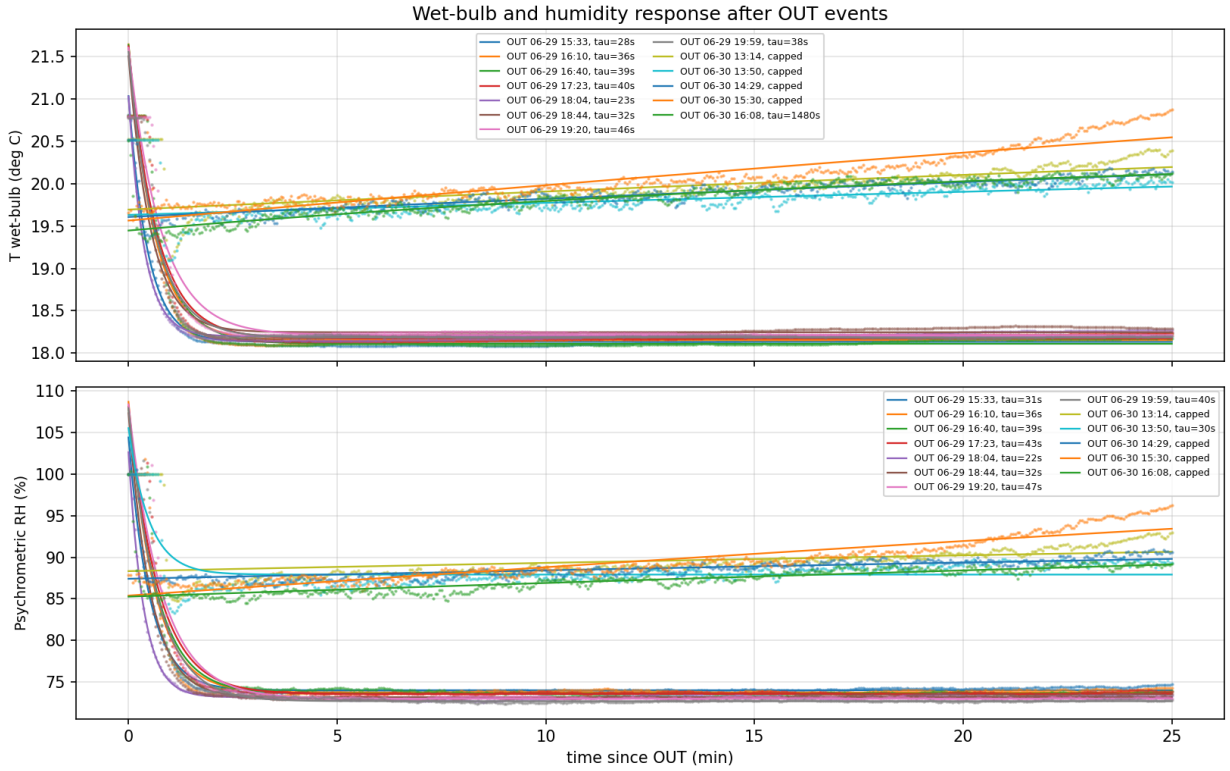


Figure 4: Wet-bulb temperature and psychrometric-RH trajectories aligned to each OUT event. Dots are observations and lines are exponential fits; labels list the OUT time and fitted time constant, or "capped" when the fit reached the upper bound. Fit windows run from OUT to min(OUT + 25 min, next IN).

Phase	N	τ_{T_d}	τ_{T_w}	τ_{dep}	τ_{RH}
2026-06-29 cotton wick	8	47	37	38	38
2026-06-30 plastic wick	5	56	$\ge 7200^*$	$\ge 7200^*$	$\ge 7200^*$

* Fit reached the 7200 s upper bound; treat those values as non-exponential or unresolved rather than physical time constants.

Phase	N	median t_{knee} (s)	median t_{63} (s)	RH at plateau (%)
2026-06-29 cotton wick	8	90	48	73.5
2026-06-30 plastic wick	5	66	30	88.5

6 Diagnostics

The full multi-panel diagnostics are still generated for traceability, but are omitted here to keep the report focused. They remain in `img/Lab20260629` and `data/20260629Lab` alongside the streamlined report figures.

7 Generated Files

- `data/20260629Lab/RBR_SHS/psychrometrics_20260629.csv`
- `data/20260629Lab/compare_SHS_tank_20260629.csv`
- `data/20260629Lab/compare_WXT_psychrometric_20260629.csv`
- `data/20260629Lab/response_time_20260629.csv`
- `data/20260629Lab/wet_bulb_settling_20260629.csv`